Name
Date
Course

Inductive and Deductive Reasoning

Inductive Reasoning: Drawing a conclusion based on a pattern taken from limited information.

Example I: My cat has hair. Your cat has hair. The next door neighbor's cat has hair. Therefore, all cats have hair.

Example II: It rained yesterday. It rained today. Therefore, it will rain every day of the year.

Example III: Old Faithful, a geyser at Yellow-stone National Park, is known to erupt many times per day since its discovery in 1870. Because this has always been true, it will continue to be true.

Conjecture: A conclusion based on inductive reasoning.

In the following problems, make a conjecture based on the information given.

① Bella likes to play the piano, the guitar, and the drums.

$\boxed{\text{Bella likes to play instruments.}}$

② Last year, Larry's tomato plant didn't produce tomatoes, nor did it produce tomatoes this year.

$\boxed{\text{Larry's tomato plant will not produce tomatoes.}}$

③ Lions, leopards, and cheetahs eat meat.

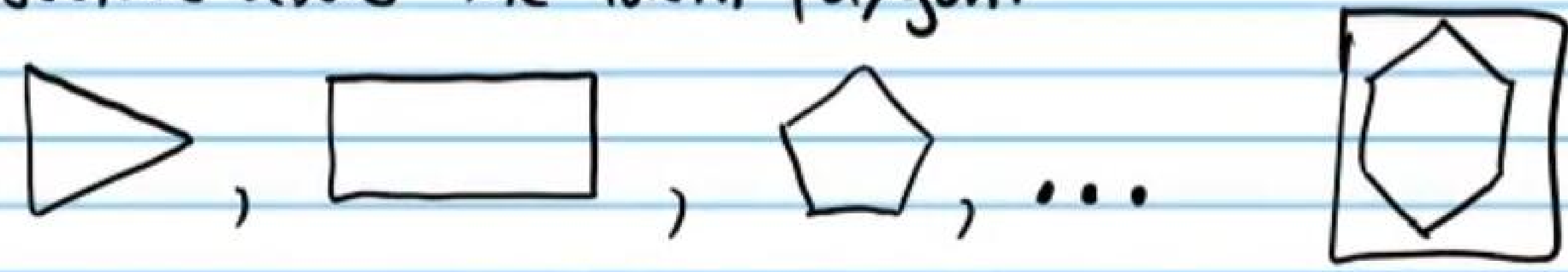
$\boxed{\text{All cats eat meat.}}$

Inductive Reasoning with Numbers and shapes

Example IV: Consider the sequence 1, 2, 4, 8 ... Use inductive reasoning to make a conjecture about the fifth number in the sequence.

$\boxed{16}$

Example V: Consider the polygons below. Make a conjecture about the fourth polygon.

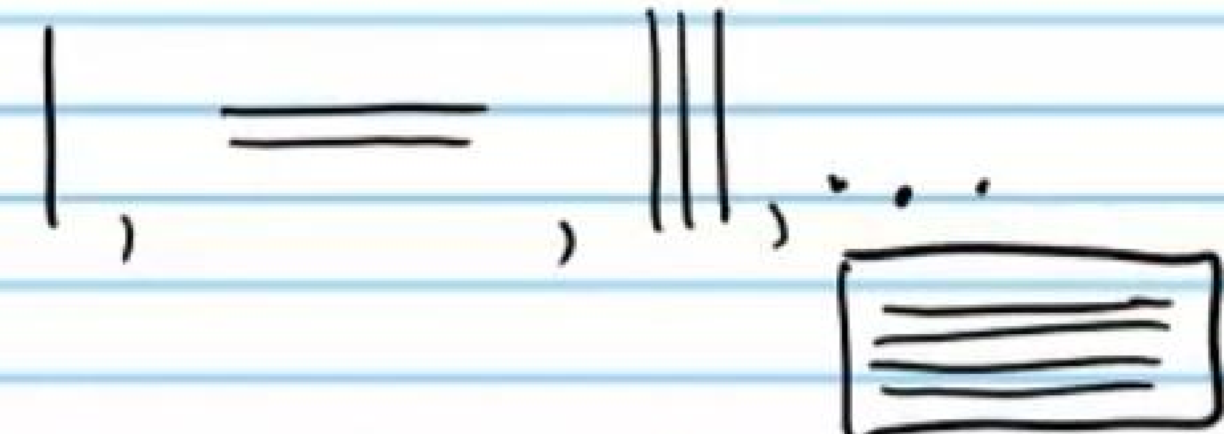


Make a conjecture about the next thing in the sequence.

④ 3, 5, 7, ...

$\boxed{9}$

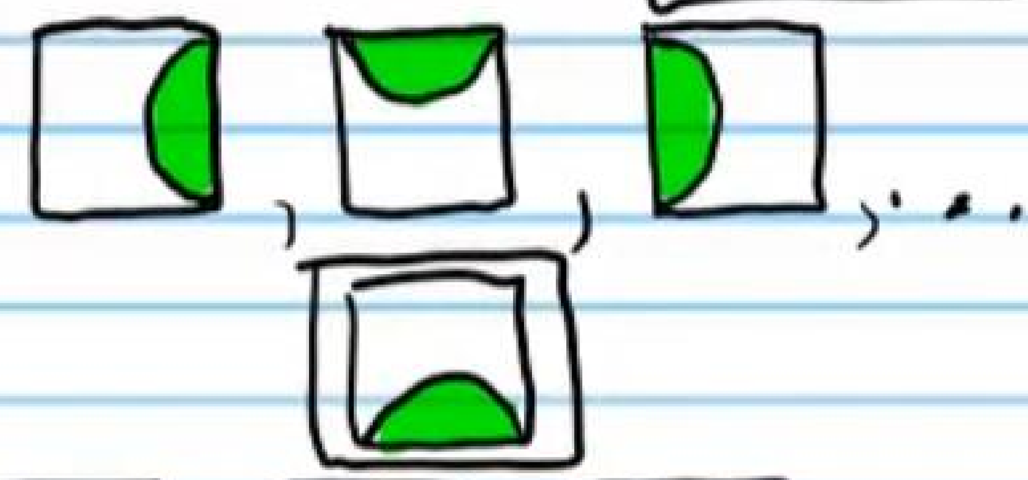
⑤



⑥ 50, 0, -50, ...

$\boxed{-100}$

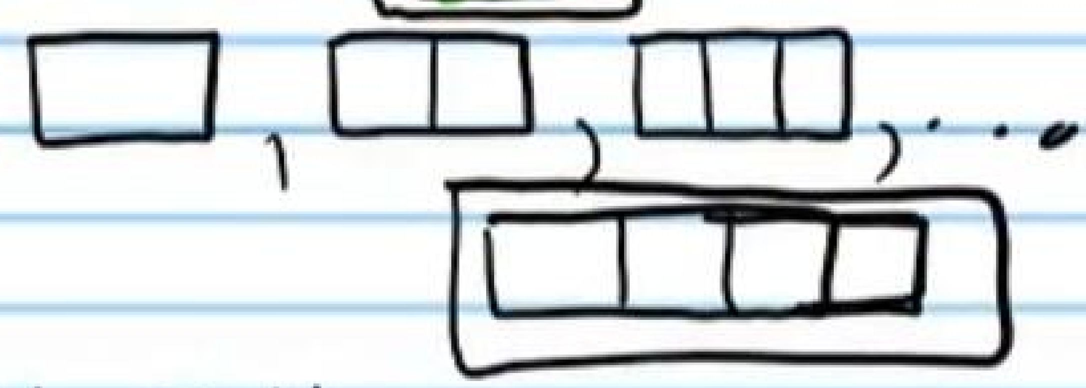
⑦



⑧ E, G, I ...

$\boxed{K}$

⑨



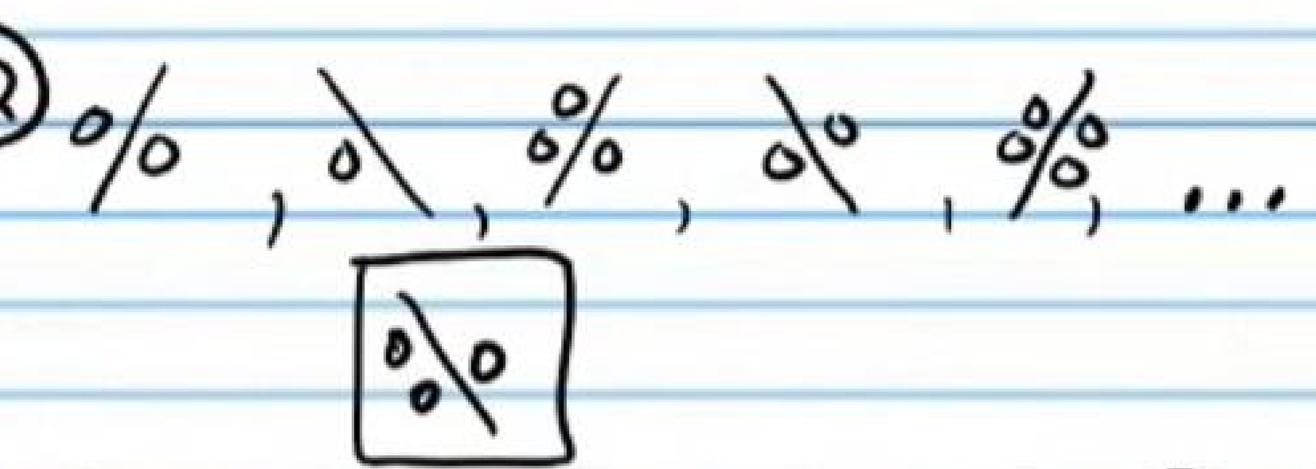
More Difficult Patterns

⑩ 1, 4, 9, 16 ...

$\boxed{25}$

⑪ 10, 9, 7, 4, ...

$\boxed{0}$

⑫ 

⑬ $5, \frac{5}{3}, \frac{5}{9}, \ldots$

$\boxed{\frac{5}{27}}$

⑭ $2, 4, 8, \ldots$

$\boxed{16}$

⑮ AB3, CD8, EF13, ...

$\boxed{\text{GH18}}$

⑯ Complete the conjecture based on the following information.
I) $2+4=6$
II) $10+30=40$
III) $16+20=36$
The sum of two even integers is <u>even</u>.

⑰ Make a conjecture based on the following information.
I) $3\times3=9$
II) $5\times11=55$
III) $7\times9=63$
The product of two odd integers is <u>odd</u>.

⑱ Complete the conjecture based on the following information.
I) $3+4=7$
II) $5+8=13$
III) $6+9=15$
The sum of an odd integer and an even integer is <u>odd</u>.

⑲ Complete the conjecture based on the following information.
I) $3\times2=6$
II) $4\times5=20$
III) $8\times7=56$
The product of an odd integer and an even integer is <u>even</u>.

Counterexample: An example that disproves a conjecture.

Find a counterexample for the following conjectures.

⑳ All cars use gasoline.

$\boxed{\text{Some cars use electricity}}$

㉑ The product of two numbers is always positive.

$\boxed{-2(3)=-6}$

㉒ The sum of two fractions is always less than $1\frac{1}{2}$.

$\boxed{\frac{8}{9}+\frac{8}{9}=\frac{16}{9}=1\frac{7}{9}}$

㉓ All parakeets are green.

$\boxed{\text{Some parakeets are blue.}}$

㉔ All prime numbers are odd.

$\boxed{2}$

㉕ x^2 is always greater than x if x is any real number.

$\boxed{\left(\frac{1}{2}\right)^2<\frac{1}{2}}$

㉖ $\sqrt{x+y}=\sqrt{x}+\sqrt{y}$

$x=1,\ y=2$ $\quad \sqrt{1+2}\neq\sqrt{1}+\sqrt{2}$

$\boxed{\sqrt{3}=1+\sqrt{2}}$

㉗ All meat comes from cows.

$\boxed{\text{Some people eat deer.}}$

Conditional Statement: A statement with a hypothesis and a conclusion.

If-Then Form: If X, then Y.

Hypothesis $\swarrow$ X   Y $\searrow$ Conclusion

Example: If it rains, then I will take my umbrella.

Write the following statements in if-then form.

㉘ $3x+6=15$ because $x=3$.

$\boxed{\text{If } 3x+6=15, \text{ then } x=3}$

㉙ All turtles have shells.

$\boxed{\text{If a creature is a turtle, then it has a shell.}}$

(30) All vertical angles are congruent.

If two angles are vertical, then they are congruent.

(31) A polygon with congruent sides is equilateral.

If a polygon has congruent sides, then it is equilateral.

(32) Identify the hypotheses and conclusions in the previous four problems.

28 →	H: $3x+6=15$	C: $x=3$
29 →	H: creature is a turtle	C: It has a shell.
30 →	H: Two angles are vertical	C: They are congruent.
31 →	H: A polygon has congruent sides	C: It's equilateral.

Negation: The opposite of an original statement.

Example I: Original Statement → The flowers are red.
New Statement → The flowers are not red.

Example II: Original Statement → The number is not even.
New Statement → The number is even.

Converse: When the hypothesis and conclusion are switched.

Original: If I feel well, then I will ride my bike.
Converse: If I ride my bike, then I will feel well.

Inverse: The hypothesis and conclusion are negated.

Original: If I feel well, then I will ride my bike.
Inverse: If I don't feel well, then I will not ride my bike.

Contrapositive: The hypothesis and conclusion are switched <u>and</u> negated.

Original: If I feel well, then I will ride my bike.
Contrapositive: If I dont ride my bike, then I will not feel well.

Write the converse, inverse, and contrapositive of the following statements.

(33) If Bill runs, then he will sweat.

Converse: If Bill sweats, then he will run. Inverse: If Bill does not run, then he will not sweat. Contrapositive: If Bill does not sweat, then he will not run.

(34) If the measure of an angle is 90°, then it is a right angle.

Converse: If an angle is a right angle, then its measure is 90°. Inverse: If the measure of an angle is not 90°, then it is not a right angle. Contrapositive: If an angle is not a right angle, then its measure is not 90°.

Equivalent Statements: Two statements that are either both true or both false.

Examples: I) A statement and its contrapositive.

Original: If a polygon is a square, then it is a quadrilateral.
Contrapositive: If a polygon is not a quadrilateral, then it is not a square.

II) The converse and inverse.

Converse: If a polygon is a quadrilateral, then it is a square.
Inverse: If a polygon is not a square, then it is not a quadrilateral.

Biconditional Statement: A statement that is true, as well as its converse. It can be written with "if and only if."

Original: If the sum of the measures of two angles is 180°, then the angles are supplementary.

Converse: If two angles are supplementary, then the sum of their measures is 180°.

These are both true, so we can combine them and write,

Two angles are supplementary if and only if the sum of their measures is 180°.

Determine whether the following statements are biconditional. If so, write in "if and only if" form. If they are not biconditional, explain why.

(35) If an animal is a dolphin, then it has flippers.

Not biconditional.

Whales have flippers

(36) If two polygons have the same size and shape, then they are congruent.

Biconditional: Two polygons are the same size and shape if and only if they are congruent.

(37) If $\frac{x}{2}+3=4$, then $x=2$.

Biconditional: $\frac{x}{2}+3=4$ if and only if $x=2$.

(38) If Bill's bicycle has a flat tire, then he cannot ride today.

Not biconditional: Bill might be busy.

(39) If $x<4$, then $x<10$.

Not biconditional: If x is 9, then it's not less than 4.

Writing Statements with Symbols

If X, then Y can be written as $X \rightarrow Y$.

Negation is written with a squiggly line ($\sim$).

Converse: $Y \rightarrow X$
Inverse: $\sim X \rightarrow \sim Y$
Contrapositive: $\sim Y \rightarrow \sim X$
Biconditional: $X \leftrightarrow Y$

Deductive Reasoning: An argument based on definitions, postulates, theorems, and logic.

Examples using Logic

The Law of Detachment: If $X \rightarrow Y$ is true, and X is true, then Y must be true.

Example

If I hurt my arm, then I will not be able to pitch in the game tomorrow.

If it is true that an injury to my arm will prevent me from pitching in the game, and if it is also true that my arm has been injured, then the conclusion, I will not be able to pitch in the game tomorrow, must also be true.

The Law of Syllogism: If $X \rightarrow Y$ is true, and $Y \rightarrow Z$ is true, then $X \rightarrow Z$ must be true.

Example

If I lose my car keys, then I will be late to the party.
If I am late to the party, then the host will be angry.

Therefore, if I lose my car keys, then the host of the party will be angry.

Determine if the Law of Detachment or the Law of Syllogism can be used with the following statements. Make a valid conclusion using one of the laws.

40. If $1-5x=11$, then $x=-2$. If $x=-2$, then Cynthia got the right answer on one of her test questions.

The Law of syllogism. | If $1-5x=11$, then Cynthia got the right answer.

41. If it rains, then I will wear my coat. If I wear my coat, then I will sweat.

Law of Syllogism: If it rains, then I will sweat.

42. If the measure of an angle is 180°, then it is a straight angle. $m\angle A$ is 180°

Law of Detachment: $\angle A$ is a straight angle

Additional Problems

Make a conjecture using inductive reasoning.

43 Daniel failed his first three physics exams.

Daniel will fail the next exam.

44 Why is it possible that your conjecture in problem 43 could be false?

Daniel could study hard and pass his next exam.

45 Find the next numbers in the sequences.

a) 5, 12, 19, ...

26

b) 152, 54, −44, ...

−142

46 Disprove the following statements with counterexamples.

a) All planes use jet engines.

Some planes use propellers.

b) Pizza is always made with pepperoni.

Some pizzas have only cheese.

c) If $x^2 = 49$, then x must be 7.

$\boxed{x = -7}$ $(-7)^2 = 49$

$(-7)(-7) = 49$

$49 = 49$

Convert the following statements to if-then form.

47 All birds have feathers.

If a creature is a bird, then it has feathers.

48 $x > 6$ because $x = 10$.

If $x = 10$, then $x > 6$.

49 Identify the hypotheses and conclusions in your if-then statements above.

H: A creature is a birds
C: It has feathers

H: $x = 10$.
C: $x > 6$

50 If a bird has wings, then it can fly.

a) Write the converse, inverse, and contrapositive of the statement above.

Converse: If a bird can fly, then it has wings.
Inverse: If a bird does not have wings, then it cannot fly.
contrapositive: If a bird cannot fly, the it doesn't have wings.

b) Determine the truth value of each statement above, including the original statement.

Original: False.
Converse: True
Inverse: True
Contrapositive: False

c) Is the original statement biconditional? Explain.

$\boxed{\text{Not biconditional: The original statement is false.}}$

51 Assign symbols to the hypothesis and conclusion of the original statement in problem # 50. Then write the converse, inverse, and contrapositive using these symbols. If the original statement were biconditional, how would you notate this with symbols?

Make a conclusion using deductive reasoning. State the law that you are using.

52 If we have no lettuce, then we will use spinach. We have no lettuce.

53 If $x^3 \leq x$, then $0 \leq x \leq 1$. x^3 is less than x.

54 If I finish my homework, then I will study for the exam. If I study for the exam, then I will get an A in my course.